

# NETWORK RECONNAISSANCE & PORT SCANNING AUDIT

Authorized Security Assessment — Kali Linux Lab

|                  |                                                                  |
|------------------|------------------------------------------------------------------|
| Prepared By      | Palla Venkata Mahesh                                             |
| Date             | 02 October 2026                                                  |
| Assessment Scope | Authorized sandbox / local lab environment                       |
| Primary Tools    | Nmap • Wireshark • Kali Linux                                    |
| Evidence Package | Nmap outputs • Wireshark capture • screenshots • findings report |

# 1. Executive Summary

This report documents the structure and execution workflow for an authorized network reconnaissance and port-scanning audit. The assessment uses Nmap to identify exposed TCP/UDP services, enumerate service versions, and perform OS fingerprinting, while Wireshark is used to inspect packet-level behavior generated during the assessment. The intended target is an explicitly authorized sandbox or lab host owned or controlled by the assessor.

For transparency, the evidence package generated with this document includes a verified local TCP listener baseline from the controlled packaging environment. Those values are real observations, but they are not represented as Nmap output. Genuine Nmap and Wireshark evidence must be collected on Kali Linux against the authorized target before the report is submitted as a completed audit.

# 2. Scope and Rules of Engagement

| Item             | Definition                                                 |
|------------------|------------------------------------------------------------|
| Tester           | Palla Venkata Mahesh                                       |
| Date             | 02 October 2026                                            |
| Platform         | Kali Linux (final execution environment)                   |
| Target           | Authorized sandbox target only                             |
| Testing          | TCP SYN, all-TCP, service/version, OS, UDP, host discovery |
| Traffic analysis | Wireshark packet capture and protocol review               |
| Restriction      | No unauthorized systems, accounts, or credentials          |

# 3. Methodology

The workflow begins with target and scope verification, followed by host discovery where a lab network is in scope. TCP SYN scanning is then used to identify open TCP ports. Service/version enumeration determines the software exposed on those ports, and OS detection provides an Nmap fingerprint estimate where supported. A UDP top-port scan identifies potentially exposed UDP services. Wireshark is run in parallel to capture and inspect the network traffic generated by the authorized assessment.

| Phase | Activity            | Evidence                |
|-------|---------------------|-------------------------|
| 1     | Scope verification  | Target IP / lab network |
| 2     | Host discovery      | host-discovery.txt      |
| 3     | TCP SYN scan        | tcp-syn.txt             |
| 4     | Full TCP scan       | all-tcp-ports.txt       |
| 5     | Service enumeration | service-version.txt     |
| 6     | OS detection        | os-detection.txt        |
| 7     | UDP scan            | udp-scan.txt            |

| Phase | Activity        | Evidence                           |
|-------|-----------------|------------------------------------|
| 8     | Packet analysis | network-recon.pcapng + screenshots |

## 4. Verified Local Baseline

The packaging environment reported the following TCP listeners on 127.0.0.1 and each listed port was independently verified with a TCP connection attempt. This is included as factual baseline data only; it is not presented as Nmap output. Verified ports: **1384, 8080, 33587, 35075, 36329, 41871, 50479, 54317**.

| Port  | Protocol | Verification          | Confirmed banner |
|-------|----------|-----------------------|------------------|
| 1384  | TCP      | TCP connect succeeded | HTTP / Uvicorn   |
| 8080  | TCP      | TCP connect succeeded | HTTP / Uvicorn   |
| 33587 | TCP      | TCP connect succeeded | Not asserted     |
| 35075 | TCP      | TCP connect succeeded | Not asserted     |
| 36329 | TCP      | TCP connect succeeded | Not asserted     |
| 41871 | TCP      | TCP connect succeeded | Not asserted     |
| 50479 | TCP      | TCP connect succeeded | Not asserted     |
| 54317 | TCP      | TCP connect succeeded | Not asserted     |

No UDP listeners were reported by the local preflight `ss -ltn` check. This does not replace an Nmap UDP scan because UDP service detection can depend on probe responses, firewalls, and filtering.

## 5. Nmap Evidence Collection on Kali

Run the supplied recon\_audit.sh script from the project root. It writes genuine Nmap output directly into the nmap-results directory. The target must be an authorized sandbox host. Replace the current preflight text files with those genuine outputs before submitting the report.

```
chmod +x recon_audit.sh

./recon_audit.sh <AUTHORIZED_TARGET_IP> <AUTHORIZED_NETWORK>
```

## 6. Wireshark Evidence

**The package contains an empty, valid pcapng container rather than fabricated packets. This is intentional: genuine packet evidence must be collected in Wireshark on Kali during the authorized Nmap scan. Save the resulting capture as wireshark-captures/network-recon.pcapng and replace the preview screenshots with original screenshots from the capture.**

Recommended display filters include tcp.flags.syn == 1 && tcp.flags.ack == 0 for SYN traffic, http for HTTP traffic, ftp for FTP, telnet for Telnet, and dns for DNS. Any observation of unencrypted credentials or sensitive data must be documented only when it occurs within the authorized lab environment.

## 7. Findings and Security Considerations

| ID   | Observation                                                    | Evidence            | Security consideration                                                     |
|------|----------------------------------------------------------------|---------------------|----------------------------------------------------------------------------|
| F-01 | Open TCP listeners exist in the controlled baseline            | local verification  | Review whether each exposed service is required.                           |
| F-02 | HTTP/Uvicorn banner was confirmed on detected HTTP listener    | Direct banner probe | Use appropriate access controls and encrypted transport where appropriate. |
| F-03 | UDP service exposure was not confirmed by the local preflight. | local preflight.    | Validate with Nmap UDP scanning on the authorized target.                  |

| ID   | Observation                                             | Evidence            | Security consideration                           |
|------|---------------------------------------------------------|---------------------|--------------------------------------------------|
| F-04 | Nmap OS fingerprint is pending final Kali execution     | Execution script    | Do not infer OS from the baseline kernel string. |
| F-05 | Wireshark packet evidence is pending final Kali capture | capture placeholder | Capture and document only authorized traffic.    |

## 8. Recommendations

- Disable services that are not required for the authorized application or lab function.
- Restrict management and database services to trusted network segments.
- Keep exposed services and operating systems patched according to the organization's maintenance process.
- Prefer encrypted protocols and secure transport for authentication and sensitive data.
- Use host and network firewalls to limit unnecessary inbound access.
- Repeat authorized reconnaissance periodically and compare results to the approved service inventory.

## 9. Conclusion

This deliverable provides a professional, auditable structure for the Network Reconnaissance & Port Scanning Audit task. The supplied automation script is ready for Kali Linux and generates the requested Nmap evidence files. The package deliberately avoids inventing Nmap output or Wireshark packets. Once the authorized Kali scan and Wireshark capture are performed, the resulting evidence can be dropped into the existing folders and the final report can be updated with the genuine observations.